

CodeAlpha Internship Project Report

Domain:	Artificial Intelligence	Academic Year:	B.Tech 2nd Year
Project Title:	Task 1 - Language Translation Tool	Submission Date:	June 2026

1. Project Overview

Yeh ek simple aur user-friendly Language Translation Tool hai jise Python aur Tkinter library ka use karke banaya gaya hai. Is project ka main objective users ko ek graphical interface provide karna hai jahan wo kisi bhi text ko ek language se doosri language mein asaani se translate kar sakein. Translation functionality ke liye isme googletrans (Google Translate API wrapper) ka use kiya gaya hai.

2. Features Implemented

- **GUI Interface:** Tkinter ka use karke ek clean aur modern UI banaya gaya hai jisme Source Text aur Translated Text ke liye alag boxes hain.
- **Language Selection Dropdown:** User Combo-box ke through easily Target language (jaise Hindi, Spanish, French, German, etc.) select kar sakta hai.
- **Real-time API Integration:** Back-end par Google Translation API call hoti hai jo accurate aur fast translation response generate karti hai.
- **Clear Button:** Interface par ek clear button diya gaya hai taaki user ek click me saara text clear karke naya translation start kar sake.

3. Python Source Code

```
import tkinter as tk
from tkinter import ttk, messagebox
from googletrans import Translator, LANGUAGES

class LanguageTranslatorApp:
    def __init__(self, root):
        self.root = root
        self.root.title("Language Translation Tool")
        self.root.geometry("550x450")
        self.root.configure(bg="#f3f4f6")

        self.translator = Translator()

        # Title Label
        title_lbl = tk.Label(root, text="AI Language Translator", font=("Arial", 16,
"bold"), bg="#f3f4f6", fg="#1e3a8a")
        title_lbl.pack(pady=10)
```

```

# Source Text Frame
src_frame = tk.LabelFrame(root, text=" Source Text (Auto Detect) ",
font=("Arial", 10, "bold"), bg="#f3f4f6")
src_frame.pack(padx=15, pady=5, fill="both", expand=True)

self.src_text = tk.Text(src_frame, height=5, font=("Arial", 11))
self.src_text.pack(padx=10, pady=5, fill="both", expand=True)

# Control Frame (Language selection and button)
ctrl_frame = tk.Frame(root, bg="#f3f4f6")
ctrl_frame.pack(padx=15, pady=10, fill="x")

lang_lbl = tk.Label(ctrl_frame, text="Select Target Language: ",
font=("Arial", 10), bg="#f3f4f6")
lang_lbl.pack(side="left", padx=5)

# Get language list for dropdown
self.lang_list = list(LANGUAGES.values())
self.lang_box = ttk.Combobox(ctrl_frame, values=self.lang_list,
state="readonly", font=("Arial", 10))
self.lang_box.pack(side="left", padx=5)
self.lang_box.set("hindi") # Default selection

self.trans_btn = tk.Button(ctrl_frame, text="Translate", font=("Arial", 10,
"bold"), bg="#2563eb", fg="white", command=self.translate_text)
self.trans_btn.pack(side="right", padx=5)

# Target Text Frame
dest_frame = tk.LabelFrame(root, text=" Translated Text ", font=("Arial", 10,
"bold"), bg="#f3f4f6")
dest_frame.pack(padx=15, pady=5, fill="both", expand=True)

self.dest_text = tk.Text(dest_frame, height=5, font=("Arial", 11),
bg="#f9fafb")
self.dest_text.pack(padx=10, pady=5, fill="both", expand=True)

# Bottom controls
clear_btn = tk.Button(root, text="Clear All", font=("Arial", 10),
bg="#ef4444", fg="white", command=self.clear_fields)
clear_btn.pack(pady=10)

def translate_text(self):
    text_to_translate = self.src_text.get("1.0", tk.END).strip()
    target_lang_name = self.lang_box.get()

    if not text_to_translate:
        messagebox.showwarning("Warning", "Please enter some text to translate!")
        return

    try:

```

```

        # Find language code from full name
        lang_code = ""
        for code, name in LANGUAGES.items():
            if name == target_lang_name:
                lang_code = code
                break

        # Perform translation
        translated = self.translator.translate(text_to_translate, dest=lang_code)

        # Output result
        self.dest_text.delete("1.0", tk.END)
        self.dest_text.insert(tk.END, translated.text)
    except Exception as e:
        messagebox.onerror("Error", f"Translation failed! Check internet
connection.
Error: {e}")

    def clear_fields(self):
        self.src_text.delete("1.0", tk.END)
        self.dest_text.delete("1.0", tk.END)

if __name__ == "__main__":
    root = tk.Tk()
    app = LanguageTranslatorApp(root)
    root.mainloop()

```

4. System Requirements & Libraries

Is tool ko run karne ke liye system par Python installed hona chahiye aur internet connection mandatory hai kyunki googletrans backend par live API queries bhejta hai. Libraries needed:

- `pip install googletrans==4.0.0-rc1`
- `tkinter` (Python ke sath standard built-in aati hai)